Practical 2

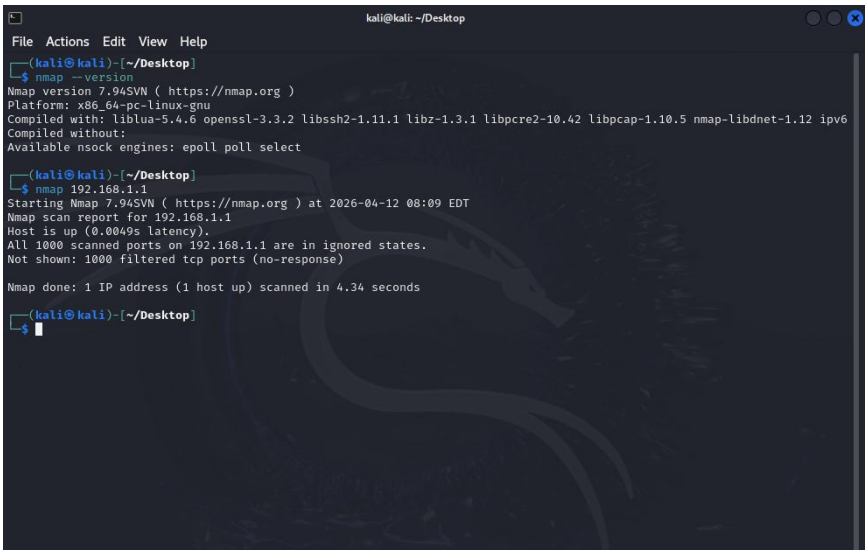
Practical No. 2: Network Scanning and Host Discovery using Nmap

Aim

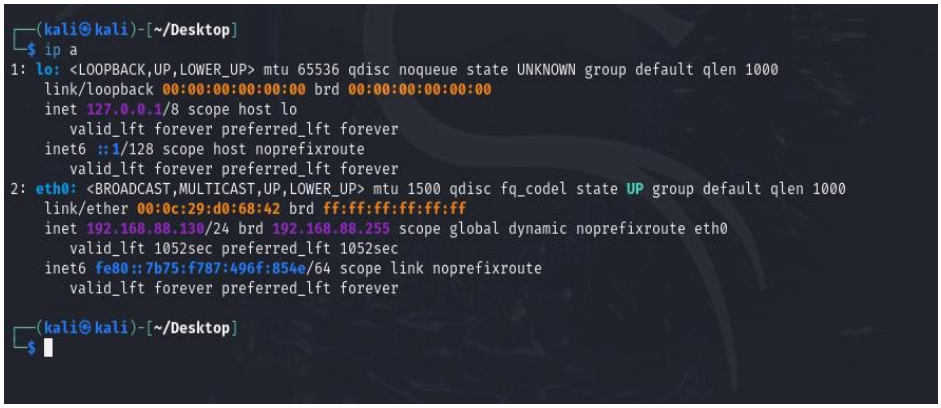
To perform network scanning and identify active hosts in a network using Nmap.

Steps-

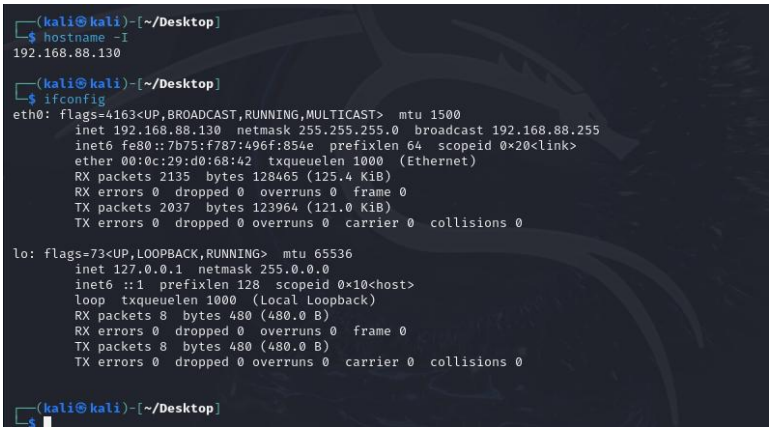
1. Step1- Launch Kali Linux terminal
2. Step2- Check Nmap Version
    - Nmap –version - Confirms that Nmap is installed and working



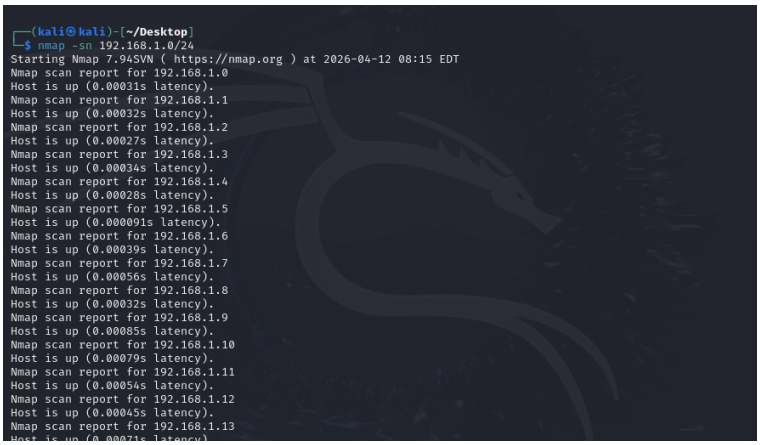
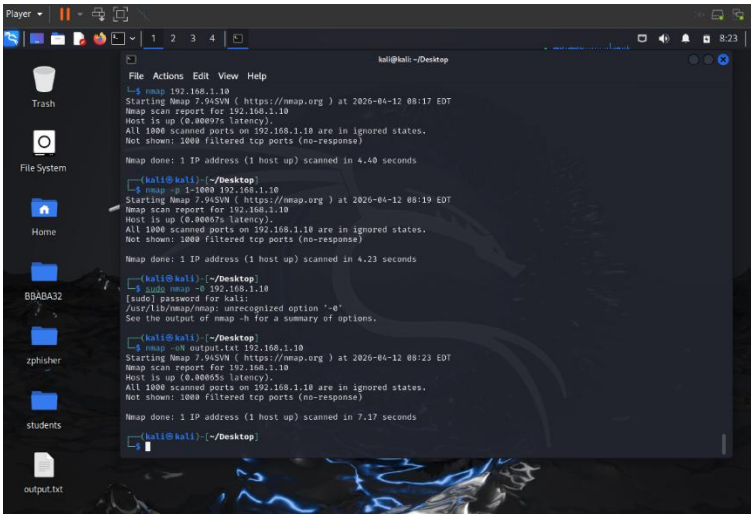
3. Step3- Check IP Address of System
    - ip a -show your IP address (like- 192.168.88.130)



4. Step4- Verify Host IP
    - hostname -I - Displays system IP address
5. Step5- Check Network Configuration
    - ifconfig - Shows detailed network interface information



6. Step6- Perform Host Discovery (Ping Scan)
    - nmap -sn 192.168.1.0/24 - Scans the entire network to find **active hosts**



7. Step7- Scan a Specific Host
    - nmap 192.168.1.10 - Scans the target system for open ports
8. Step8- Scan Specific Port Range
    - nmap -p 1-1000 192.168.1.10 - Scans ports from 1 to 1000
9. Step9- Save Scan Results to File
    - nmap -oN output.txt 192.168.1.10 - Saves scan output into a file named **output.txt**